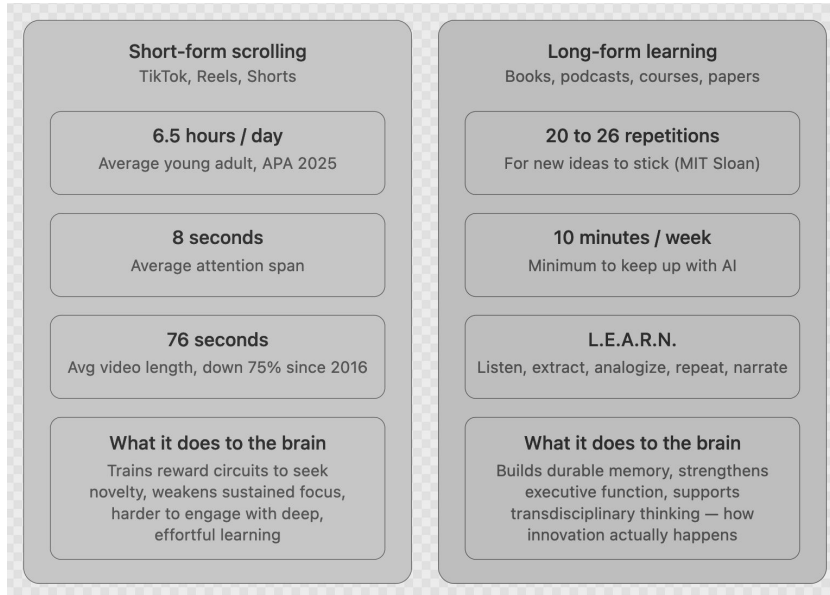


Fine-Tuning Transformers

Module 7 Week A – Adapting Language Models to Tasks

Get to know Lana — Fun Fact

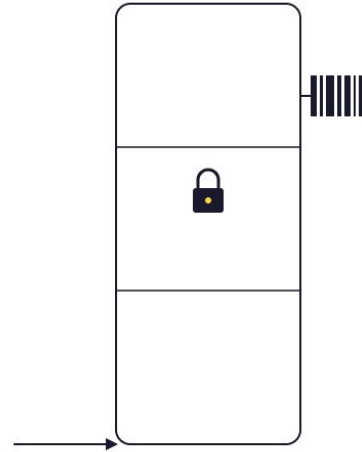
AI.SPIRE



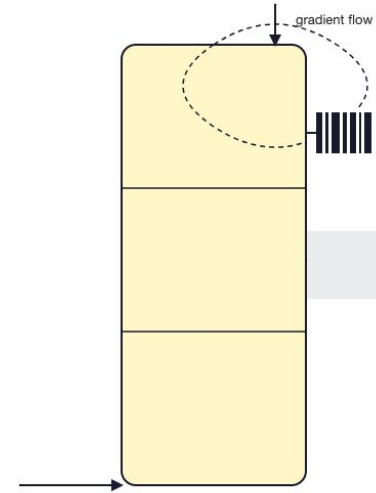
I love to learn

- Long form over short form
- Brain science & cognition
- Thought leaders, not influencers
- Continuous learning > entertainment
- Some entertainment, but with intention

From Locked To Updated



M6 Week B



M7 Week A

2014 to 2026

2014



50,000
examples



design
features



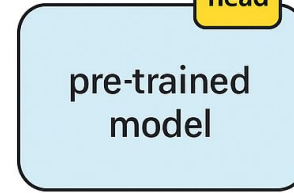
1 week



1 classifier

Recent Years

head



3,000
examples

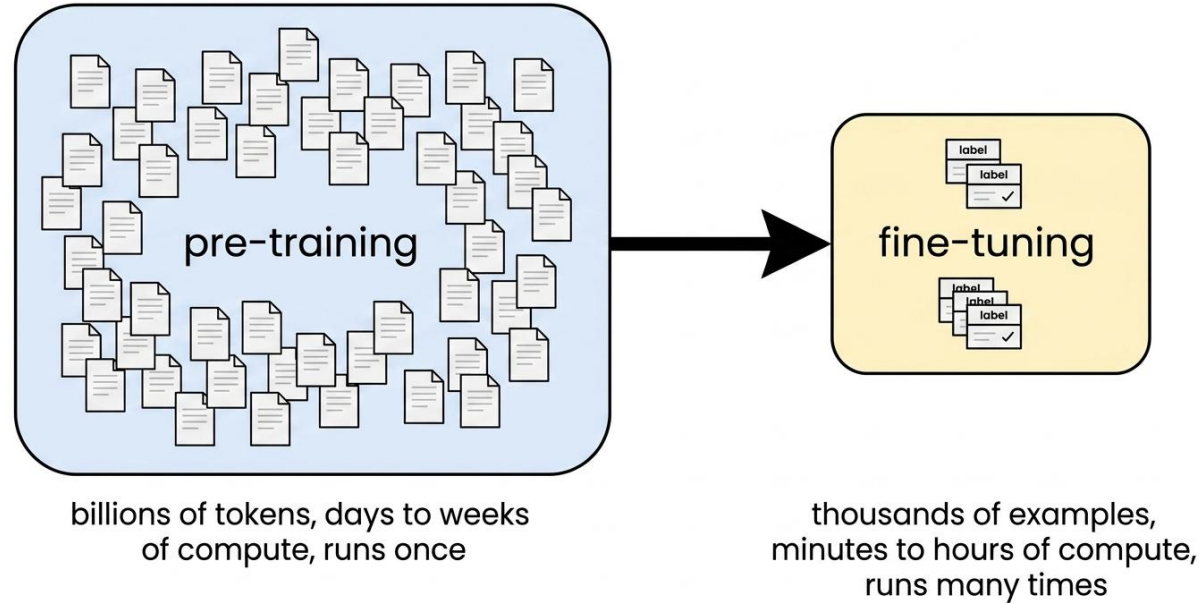


30 minutes

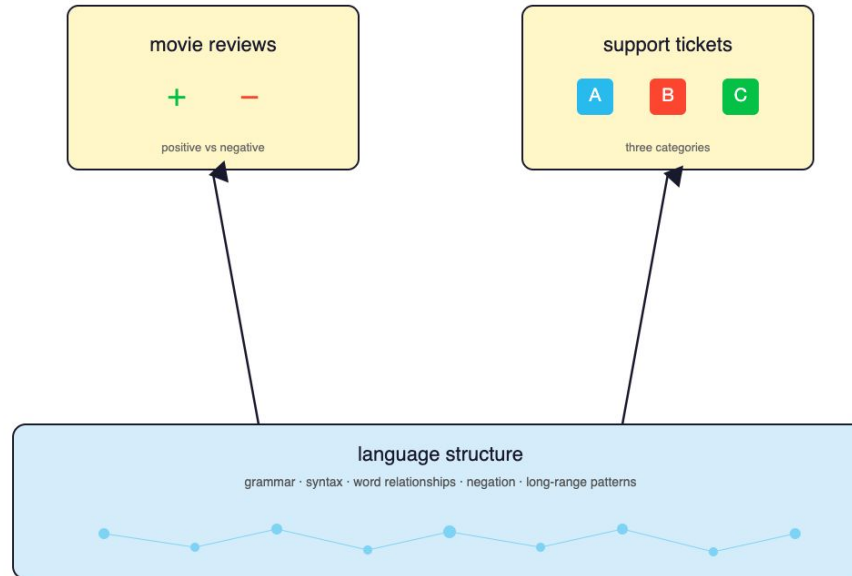


1 classifier

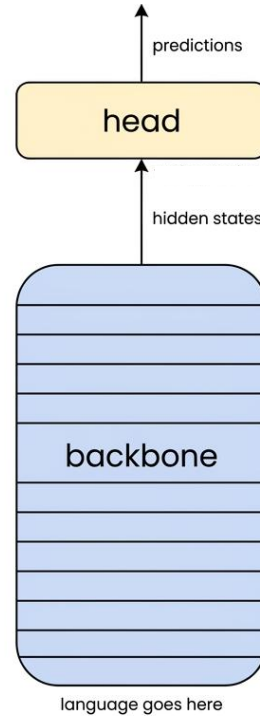
Pre-Train, Then Adapt



Language Is Reusable



Backbone Plus Head



Cost Comparison

pre-training BERT



tens of thousands of dollars, days of compute

fine-tuning DistilBERT

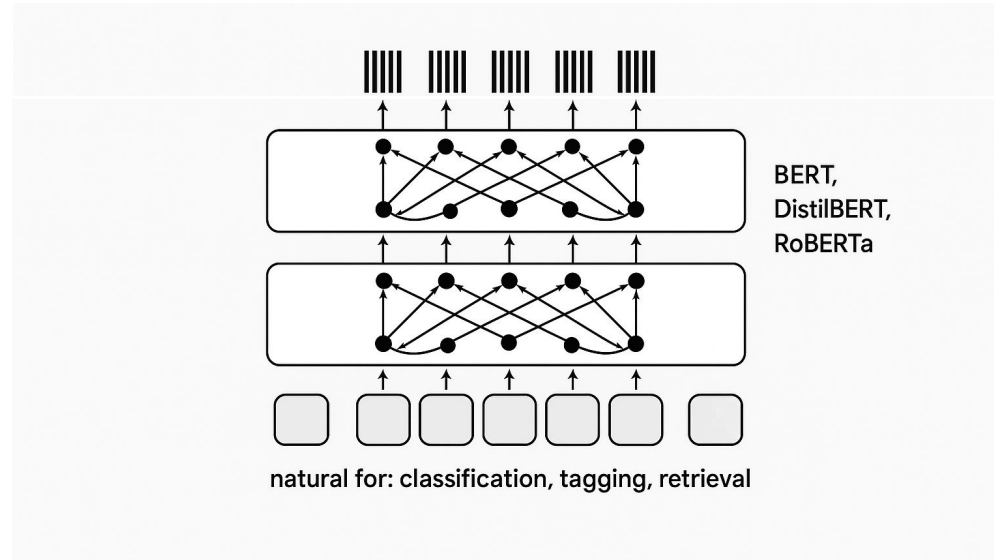


thirty minutes on a laptop

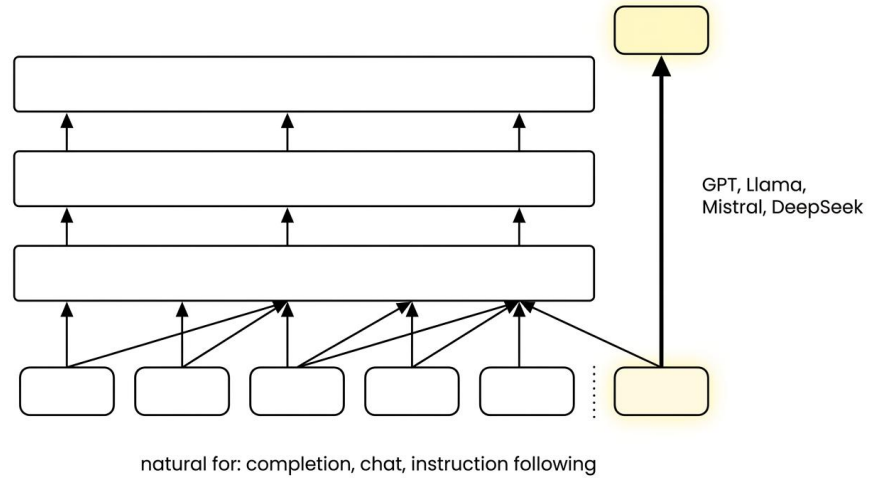
Three Families, Three Jobs

Encoder. Decoder. Encoder-decoder.

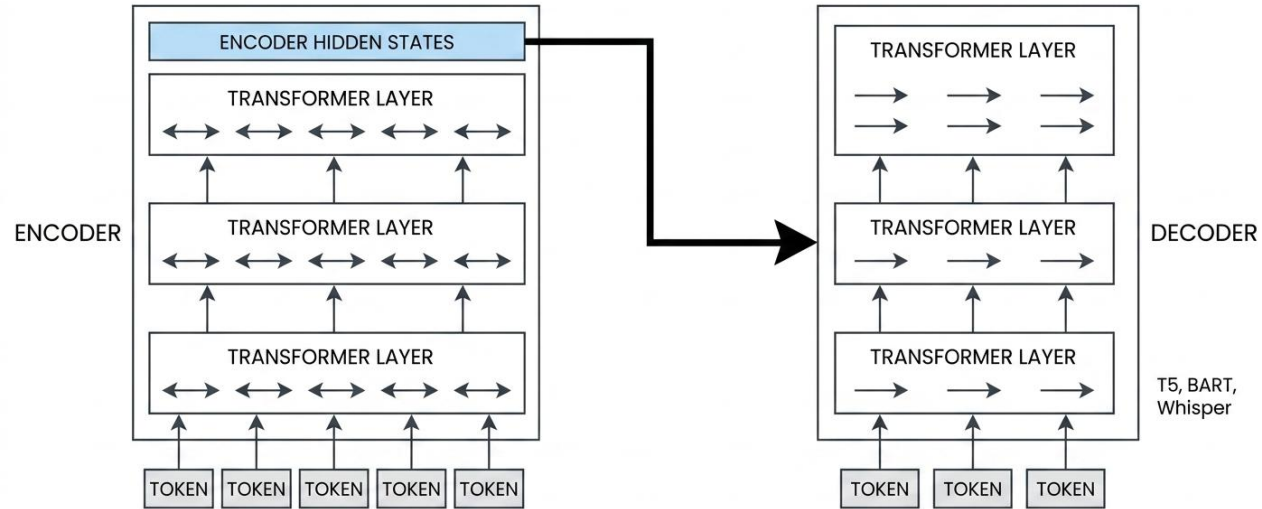
Encoder



Decoder

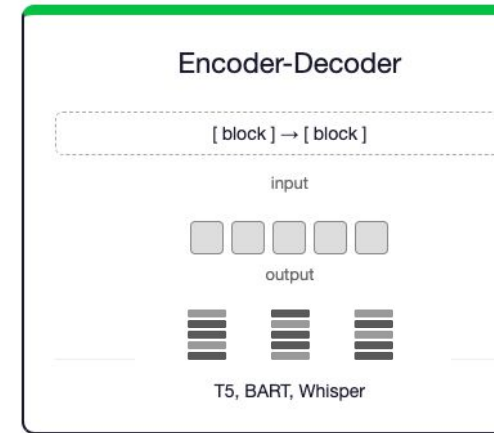
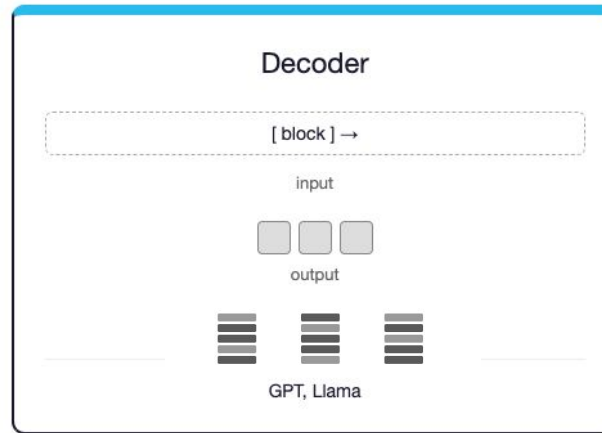
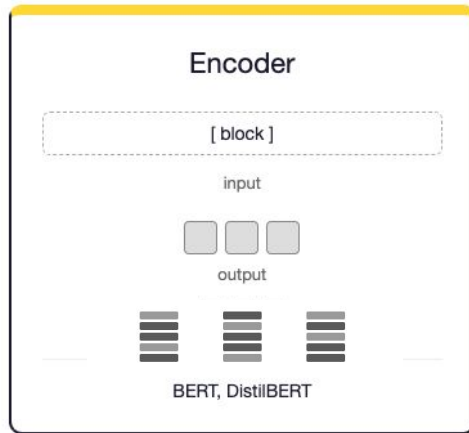


Encoder-Decoder

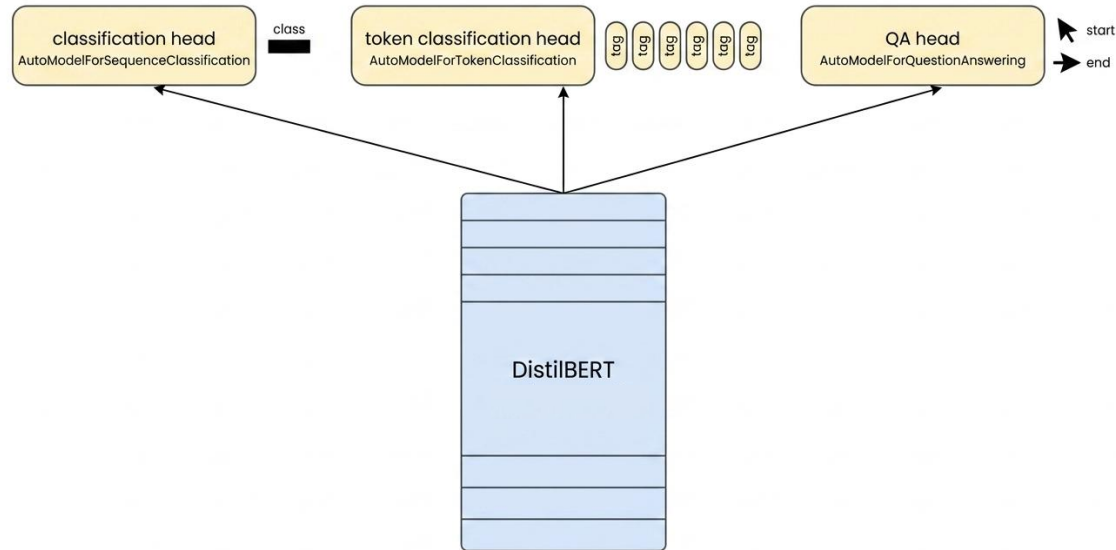


natural for: translation, summarization, transcription

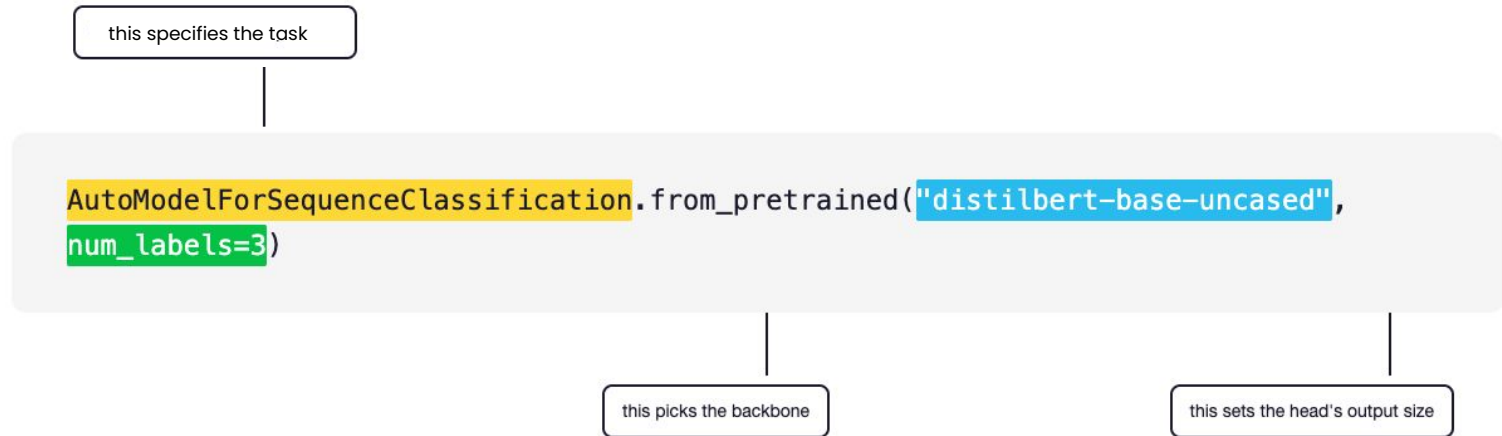
Three Families Side By Side



Same Encoder, Many Tasks



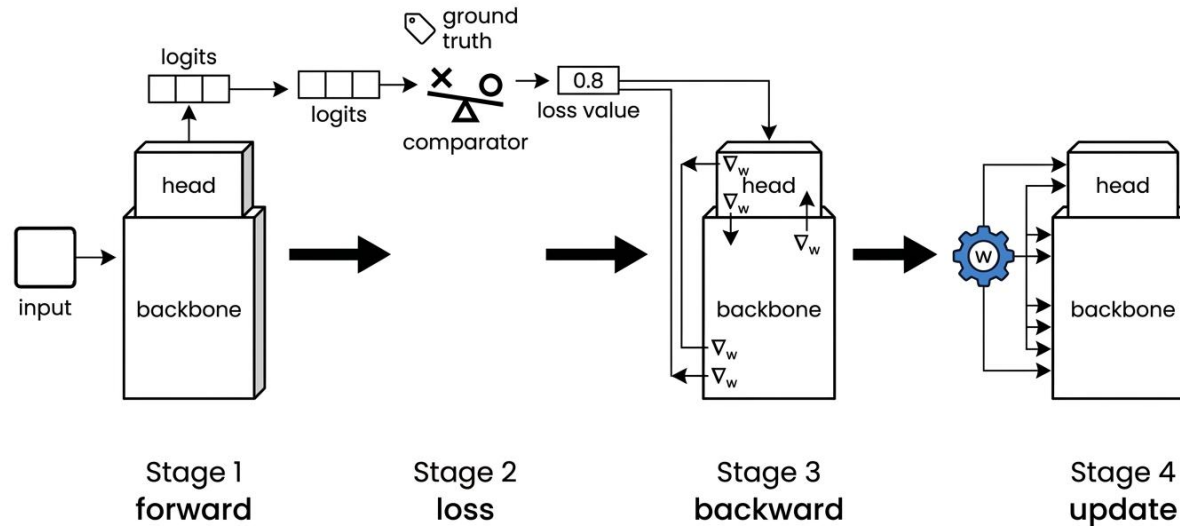
The Loading Line



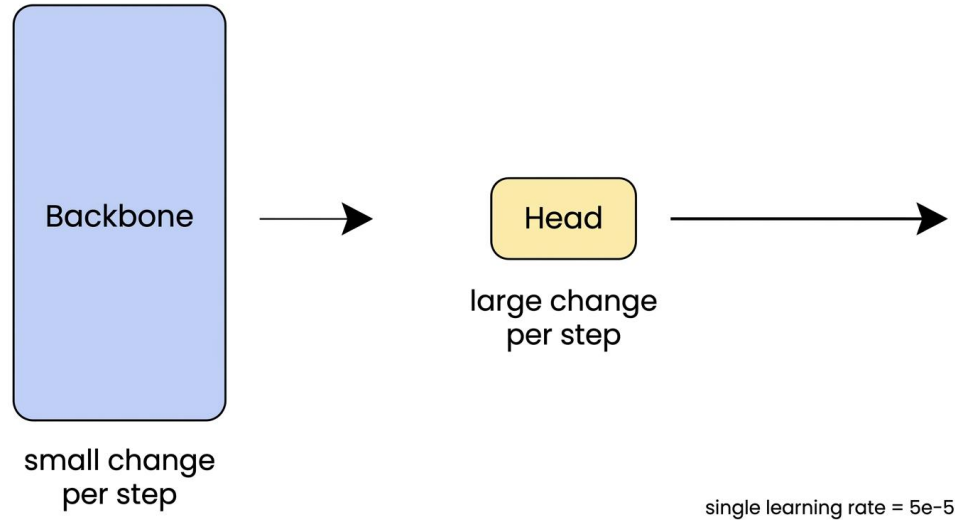
One step. Forward, loss, backward, update.

What Fine-Tuning Does

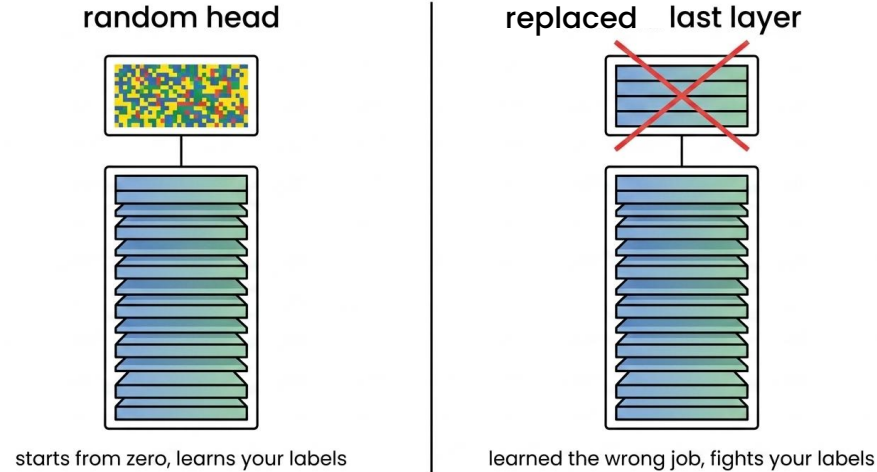
One Training Step



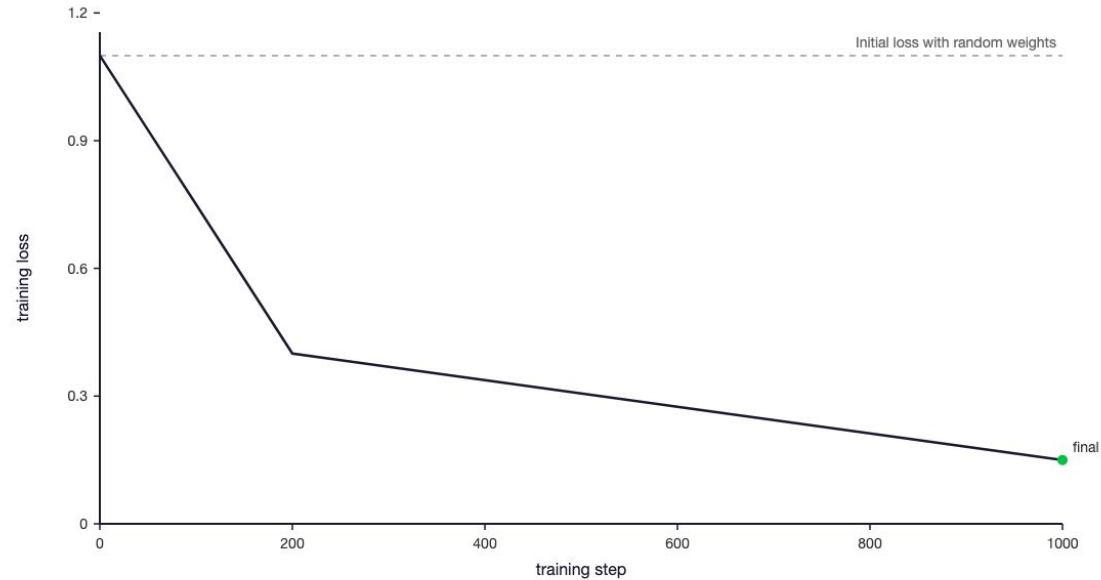
Learning Rate Asymmetry



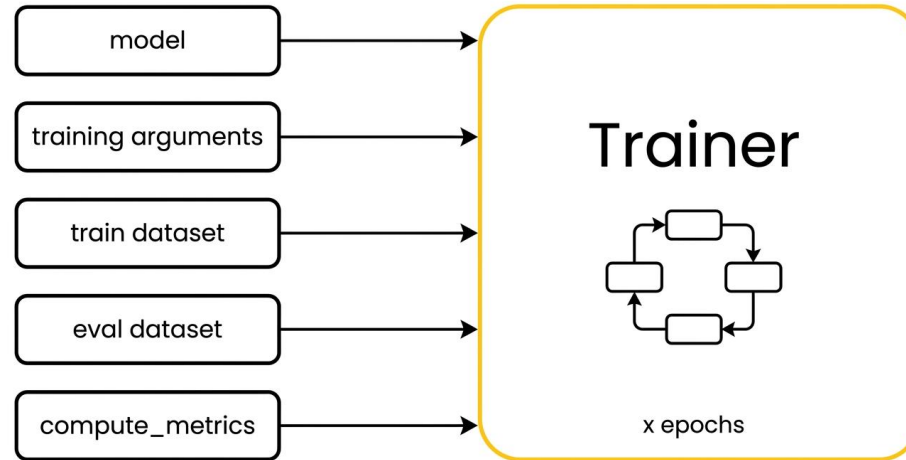
Why Random Heads



Loss Goes Down



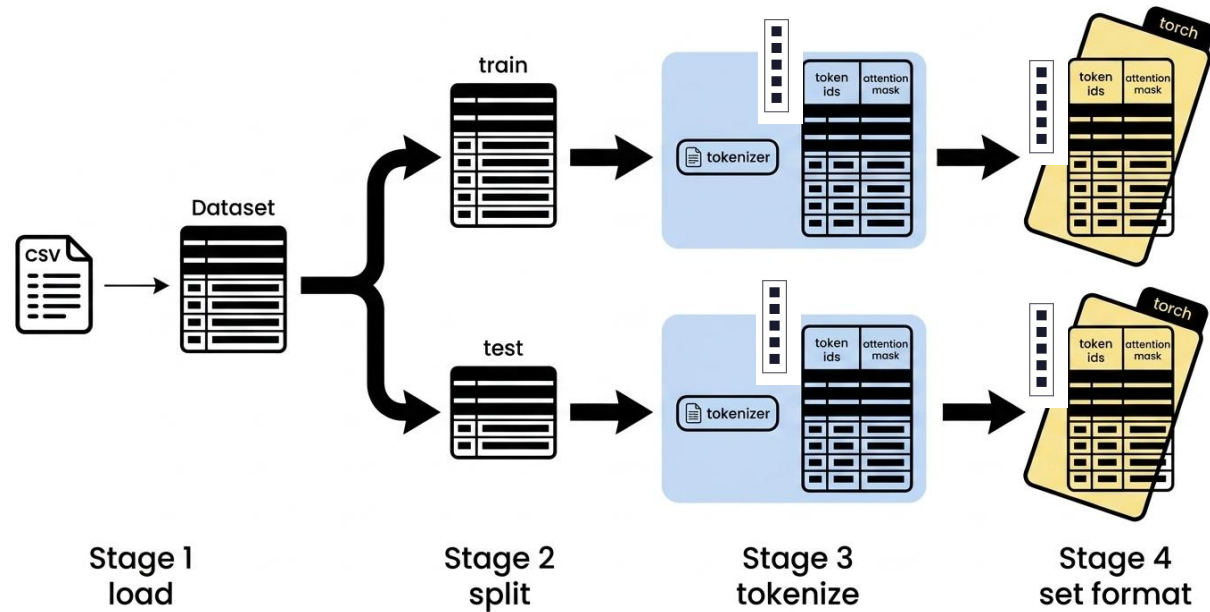
Trainer Does The Loop



Hyperparameter Choices

parameter	lab value	what it controls
learning rate	5e-5	size of each weight update
epochs	2	full passes through the training data
batch size	8	examples per training step
max length	128	maximum tokens per example

Dataset Prep In Four Steps



Compute Reality

DistilBERT (66M)



approximately 30 minutes on a laptop

BERT-large (340M)



approximately 4-6 hours on a laptop (8-10x DistilBERT)

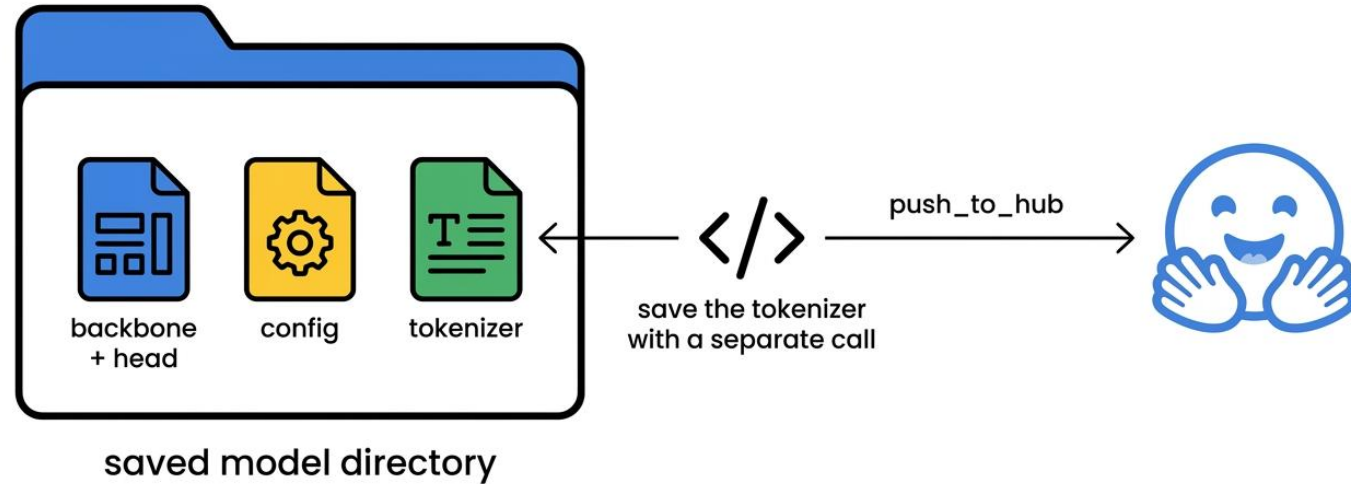
Llama (7B)



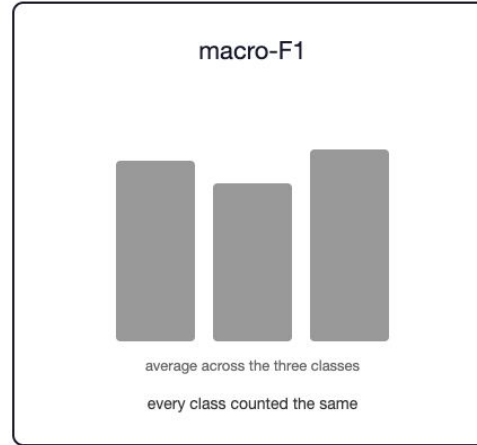
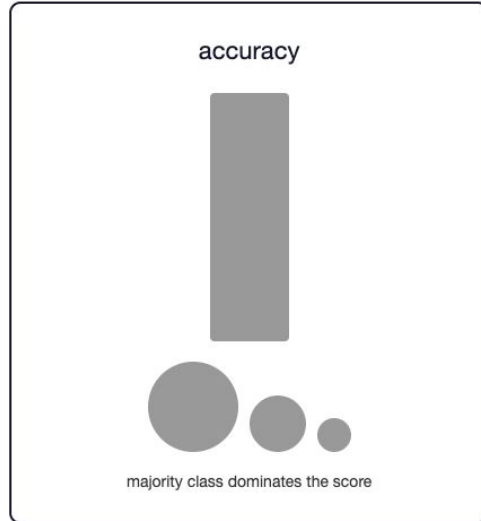
does not fit in laptop memory ▶▶▶

same data, same task

What Gets Saved



Macro-F1 Is Your Headline

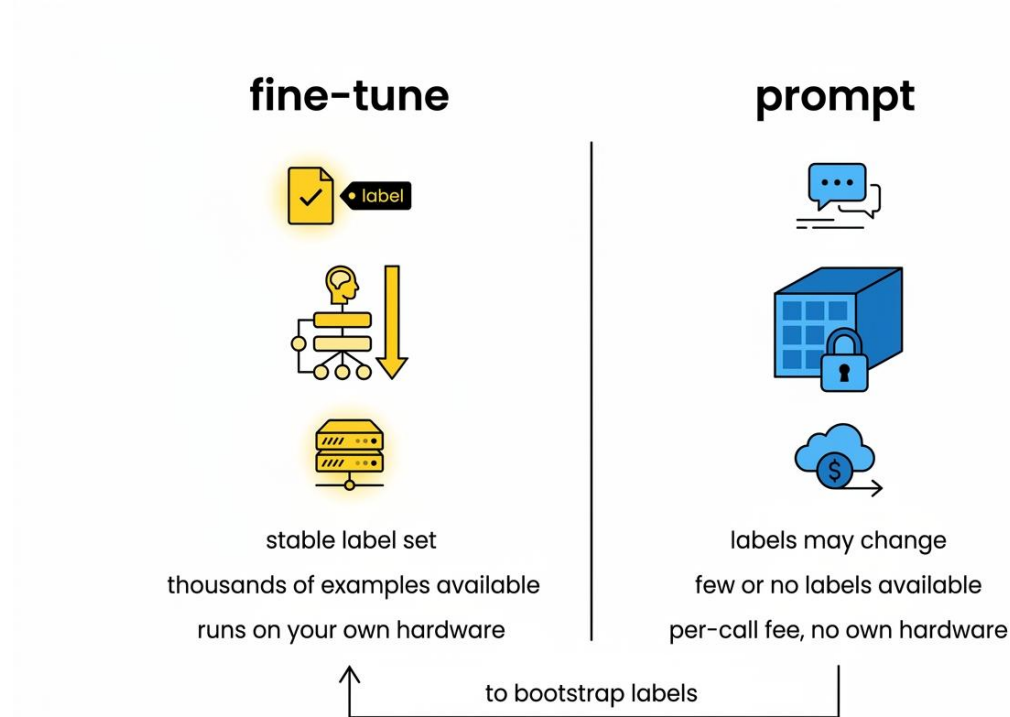


confusion matrix

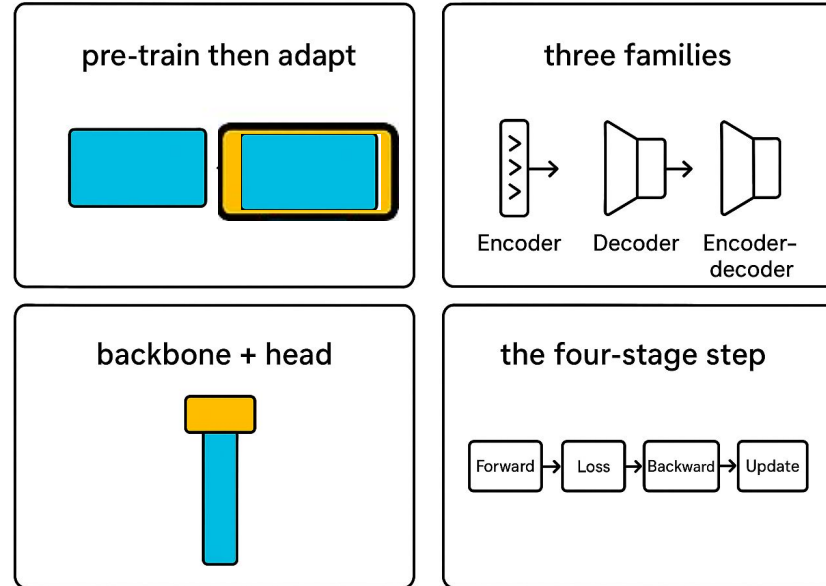
	pos	neu	neg
pos	142	8	3
neu	11	76	9
neg	4	7	128

true class (rows) vs predicted (cols)

Fine-Tune Or Prompt



What You Walked Away With



Questions?

Announcements

Congratulations!



This week's Module Mentors

- Nayef
- Hosam
- ?
- ?